

Day 7 Summary — ED Pathway + Embedded Mesh

Run status: agents ran clean; trainer + mesh integration OK.

patients	(300, 10)	ems	(300, 9)
labs	(600, 4)	transfers	(300, 6)
final units	ED-BOARD, EDObs, ICU, StepDown, Ward		
split	train/val/test = 210/45/45	pos_rate_train	0.148
pos_weight	5.77	params	26817

Validation by epoch

epoch	train_loss	val_auc	val_bce
1	1.2860	0.695	0.709
2	1.1857	0.741	0.709
3	1.2141	0.748	0.704
4	1.2094	0.774	0.705
5	1.1792	0.786	0.697
6	1.2484	0.797	0.687
7	1.1340	0.805	0.678
8	1.1272	0.816	0.668

Test set (n=45): AUC=0.564, BCE=0.697

Interpretation: Validation improved to ~0.82 by epoch 8, but test AUC ~0.56 on n=45 → high variance / possible distribution gap. Move to repeated stratified CV; add calibration and regularization.

Disposition / Flow (test snapshot)

n=45; icu_req_rate=0.00; admit_rate=1.00; board_rate=0.44; regional_rate=0.27
final_units: EDObs:24, ED-BOARD:20, StepDown:1

Signals: ICU requests did not fire in test while ~44% boarded. Instrument ICU request preconditions and capacity at the decision point; print 5 example traces.

What to fix next (short)

- Evaluation: 5x repeated stratified CV or bootstrap CIs; keep larger holdout if possible.
- Thresholding: add calibration; track prob at fixed sens/spec.
- Regularization: small dropout/L2; early stop on val AUC; checkpoint best epoch.
- Flow realism: enable ICU request criteria + 30-min retry loop.
- Troponin: switch to Roche hs-cTnT 0/1-h with CKD guard; schedule 1h ECG+troponin.
- Audit: write JSONL per action (who/when/why).
- Artifact dir: /mnt/data/ed_trainer_v13_3_run_20250811-150810